KRITI YADAV

Senior Forward Deployed AI Engineer | Agentic Systems | LLM Production Delivery | Data Science | Azure | Melbourne, Australia

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Graduate Visa (485) from July 2026 full unrestricted working rights

PROFESSIONAL SUMMARY

Production AI engineer and builder with 4.5+ years of end-to-end delivery ownership across agentic AI systems, LLM pipelines, and enterprise ML operating with a high-agency, founder's mindset from problem discovery through to deployed, iterated product. I don't build prototypes and hand them off: I own the full arc scoping, system design, build, deployment, observability, and the field feedback loop that improves the product. Shipped a multi-component agentic governance platform on GCP (Gemini 2.0, Vertex AI, RAG), a production multi-label NLP classifier on Azure ML (DistilBERT, GPT-4o orchestration, automated retraining), and an LLM evaluation harness across 1,055+ tasks with 8-layer validation and failure-recovery infrastructure. Equally fluent co-building with customer engineering teams and translating field complexity integration blockers, data readiness issues, state management problems into reusable production patterns. Completing Master of Data Science at RMIT (June 2026); research focused on LLM evaluation design and prompt architecture.

TECHNICAL SKILLS

Agentic AI & LLM Systems: Agentic workflow design and orchestration | Tool use and multi-step AI workflow architecture | RAG pipeline design and production deployment | Prompt engineering and context/memory management | LLM evaluation harness design | Observability frameworks (confidence monitoring, per-tag threshold tuning, anomaly alerting) | Hugging Face Transformers (DistilBERT, fine-tuning) | Gemini 2.0 | Vertex AI | OpenAI API (GPT-4o) | LangChain | Weak supervision and label validation at scale

Cloud, MLOps & Full-Stack: GCP (Vertex AI, Firestore) | Azure ML (deploy, monitor, retrain, API integration) | AWS (Bedrock, SageMaker, security architecture) | FastAPI | Next.js | REST API and external system integration | Docker | MLflow | CI/CD pipelines | ETL architecture (PySpark, SQL) | Audit-trail and compliance frameworks

Languages & Engineering Fundamentals: Python (pandas, scikit-learn, PyTorch) | JavaScript/TypeScript | Advanced SQL (PostgreSQL, Databricks, Redshift) | Data structures and system architecture | XGBoost | LSTM/RNN | Statistical inference (ANOVA, chi-square, Fisher exact) | A/B testing | Power BI (Master Certified) | BigQuery | Snowflake

SHIPPED AI PRODUCTS

CGAI Enterprise AI Governance & Compliance Platform — 2026

Sole Owner: Product | Architect | Builder | GCP | Gemini 2.0 | Vertex AI | RAG | Next.js | FastAPI | github.com/Kriti-Data-Business/CGAI| Pitch

- Conceived, scoped, and shipped a production agentic AI governance platform on GCP acting as sole owner across product, engineering, and deployment. Built a multi-tenant RAG pipeline over APRA/OAIC/ACMA regulatory corpora using Gemini 2.0 and Vertex AI: document ingestion agents, policy-check orchestration, guardrail enforcement, per-request audit evidence generation, and an interactive compliance copilot. Solved legacy document integration, security perimeter design, state management across multi-step agent interactions, and observable, explainable AI output end-to-end.
- Architected the full connective tissue between Vertex AI, Firestore, Cloud Run, and a Next.js front-end integrating disparate regulatory data sources across tenants, enforcing per-request traceability, and codifying the compliance-check agent into a reusable module deployable across any APRA-regulated enterprise stack.

TIO Complaint Intelligence Pipeline — 2026

Production | NLP | DistilBERT | Azure ML | LLM orchestration | Agentic retraining loop | Demo: kriti-data-business.github.io/portfolio-data-science/Complaint_Tagging_Pipeline_Interactive_Demo.html

- Designed and built a production-grade multi-label agentic classification system (DistilBERT + OvR, 90+ regulatory categories, 140k+ complaint corpus) deployed on Azure ML with full API integration zero handoff to the data team. Used GPT-4o as an orchestrated agent for weak supervision, label validation at scale, and qualitative error analysis across ambiguous complaint categories: practical multi-step LLM orchestration interacting with external data systems in a compliance-critical environment.
- Engineered the observability and iteration layer: per-tag confidence-score monitoring, automated anomaly alerting when live complaint distributions drifted, QA pipelines, and an automated retraining feedback loop. Designed a Power BI dashboard surfacing AI-driven triage signals to operations leadership. Validated both technical performance and business value to senior stakeholders.
- Identified a repeatable field pattern during deployment: inter-officer tagging inconsistency (188 raw tag variants across 140k cases) caused silent label noise that standard pipelines would miss. Designed a CleanLab-based label quality layer to surface and resolve this a reusable data readiness module now applicable to any multi-labeller compliance annotation pipeline.

LLM Evaluation Harness Code-Centric IR — 2025- Present

RMIT Research | 1,055+ tasks | 3 prompt strategies | 2 platforms | 3 difficulty tiers | github.com/Kriti-Data-Business/Research-LLM-Code-Eval-Strategy-Competitive-Programming

- Designed and built a two-phase production-grade LLM benchmarking framework: 8-layer output validation, automated failure detection, iterative rerun mechanisms, append-safe multi-day infrastructure. Codified repeatable evaluation patterns into reusable modules. Confirmed prompt strategy explains near-zero variance across all difficulty tiers; surfaced a platform × difficulty interaction unreported by LiveCodeBench.

- Made deliberate trade-offs between scope, speed, and infrastructure robustness across a 1033- to 450-problem controlled experiment balancing statistical rigour (three-way ANOVA, chi-square, Fisher exact) with delivery across multi-day compute constraints.

PROFESSIONAL EXPERIENCE

Data Scientist & AI Engineer (Intern) **Mar 2026 - Jun 2026**

Telecommunications Industry Ombudsman (TIO) | Melbourne | Government | Production AI Delivery

- Embedded directly with the DA team as the sole AI engineer: owned problem discovery, data readiness assessment, solution scoping, build, and production deployment across a 12-week cycle with no external handoff.
- Identified and resolved the core data readiness blocker (inter-officer tagging inconsistency across 188 tag variants) that would have silently degraded any classifier designed a CleanLab-based label quality framework as a reusable resolution pattern. Scoped and sequenced delivery to protect timeline while absorbing this mid-project scope change.
- Built automated retraining, anomaly alerting, and QA pipelines into production from day one not as an afterthought. Designed stakeholder observability dashboard and validated technical performance and business value to operations and senior leadership throughout delivery.

AI Research Engineer **Jun 2025 - Present**

RMIT University | Melbourne | LLM Evaluation & Prompt Architecture | Targeted: SGCI Workshop

- Designed a controlled multi-variant LLM evaluation framework across 1,055+ tasks built as production infrastructure, not a research script. 8-layer validation, automated failure recovery, append-safe resumption across multi-day runs. Codified evaluation patterns into reusable modules.
- Confirmed prompt strategy explains near-zero variance across all difficulty tiers a finding with direct implications for teams over-investing in prompt optimisation without evaluation infrastructure. Translating findings into generalised design principles and reusable evaluation playbooks.

Senior Data Scientist & AI Engineer **May 2022- Jul 2024**

Axtria Inc | Global Pharma & Healthcare | Fortune 500 clients | US/India/APAC delivery

- Designed, built, and deployed an agentic Next Best Action recommendation engine (XGBoost, LSTM/RNN, Reinforcement Learning) that orchestrated multi-step customer journey sequences replacing static promotion rules with journey-aware personalisation that made tool-use decisions across brand and channel contexts. 8.3% revenue uplift across brand- and region-specific pipelines. End-to-end ownership from design through C-suite production sign-off.
- Co-built with client engineering teams across US, India, and APAC to embed production ML best practices transferring engineering standards, not just shipping code. Designed and delivered SQL and ML training curriculum for 100+ analysts.
- Engineered PySpark + SQL ETL pipelines processing millions of daily records (66% latency reduction, zero critical data breaches across 12+ concurrent projects); maintained AI-driven reporting infrastructure for 500+ users across 20+ reports. Made trade-offs between speed and maintainability across concurrent delivery against tight client timelines.

Automation & AI Analyst **Sep 2020 - May 2022**

Wunderman Thompson MSC (WPP) | Ford (global e-commerce) & Pfizer (7 countries)

- Built automated ETL pipelines and analytical infrastructure integrating web analytics, CRM, ad spend, and e-commerce data across Ford's global footprint and Pfizer's 7-country rollouts 90% reduction in manual workflows, >60% error reduction, >99% data accuracy. Delivered A/B testing, segmentation, and purchasing behaviour insights that directly influenced go-to-market and media investment decisions for C-level stakeholders.

EDUCATION

Master of Data Science **Jul 2024 - Jun 2026**

RMIT University | Melbourne | High Distinctions: Computational ML, Applied Analytics, Data Wrangling

Bachelor of Technology Computer Science **2016 - 2020**

The NorthCap University | India

ACHIEVEMENTS & CERTIFICATIONS

- GDG Melbourne x Google DeepMind AI Sprint Top 9 team: Local Pulse dynamic intelligence for Australia's 2.4 million small businesses using Google Ai studio| Demo
- RMIT GenAI Hackathon Best Team Melbourne | Top 3 Globally: Shipped production-ready AI platform (FastAPI + AWS Bedrock + Hugging Face) with AES-GCM encryption, RBAC, adversarial testing, prompt injection vulnerability pipelines, and real-time monitoring end-to-end ownership under competitive conditions with ambiguous requirements.
- SQL and ML Trainer, Axtria: Designed and delivered training curriculum for 100+ analysts engineering enablement at scale.
- Certifications: Power BI Master Certified | AWS Machine Learning & Decision Making | UiPath RPA Developer | Advanced MySQL | Stanford Algorithms | Project Management | Python Automation

REFERENCES

Available on request Jonathan Kaija (Lead Data Analyst, TIO) | Dhivya Kaliappan (Manager Data Analytics, TIO) | Lida Rashidi & Falk Scholer (RMIT) | Hemant Kumar, Senior Director (Axtria)